

Forearm

Department of anatomy

Musculoskeletal System

First Part

The Upper Limb (6)

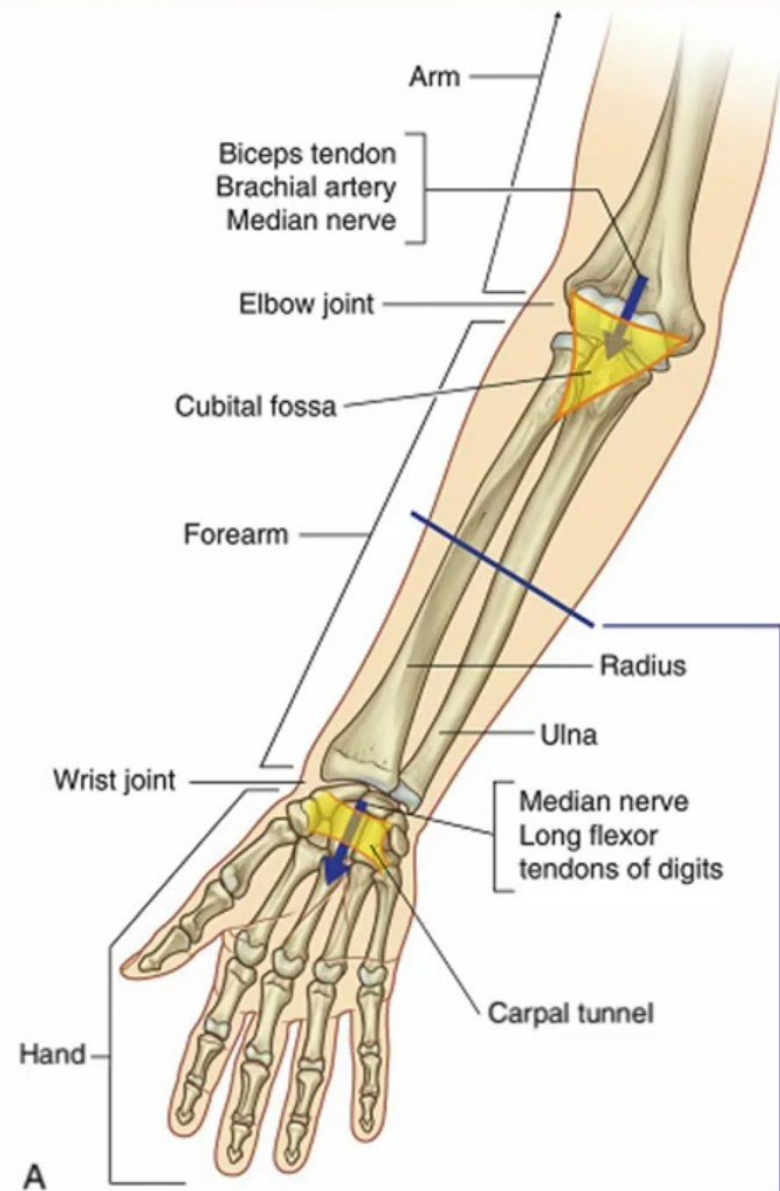
Dr. Osman Ahmed Osman

Lecture points

- ✓ Introduction
- ✓ Compartments.
- Muscles of the Forearm
- ✓ Contents of the Anterior Fascial Compartment of the Forearm
 - ✓ Muscles.
- ✓ Contents of the Posterior Fascial Compartment of the Forearm.
 - ✓ Muscles.
- Blood supply in the forearm:
 - Ulnar artery in the forearm.
 - Anterior interosseous artery.
 - Posterior interosseous artery.
 - Radial artery in the forearm.
 - Surface Marking of ulnar and radial artery.
- ✓ Summary of arteries of the Anterior and Posterior Compartment of the Forearm
- ✓ Venous drainage in the arm and Forearm
- ✓ Lymphatic drainage in the forearm
- Innervation in the Forearm.
- ✓ Median nerve in the anterior fascial compartment of the forearm.
- ✓ Anterior Interosseous Nerve .
- ✓ Ulnar nerve in the anterior fascial compartment of the forearm.
- ✓ Radial Nerve and the posterior interosseous nerve of the Forearm
- ✓ Innervation :
 - ✓ Sensory (dermatome).
 - ✓ Motor (myotome).

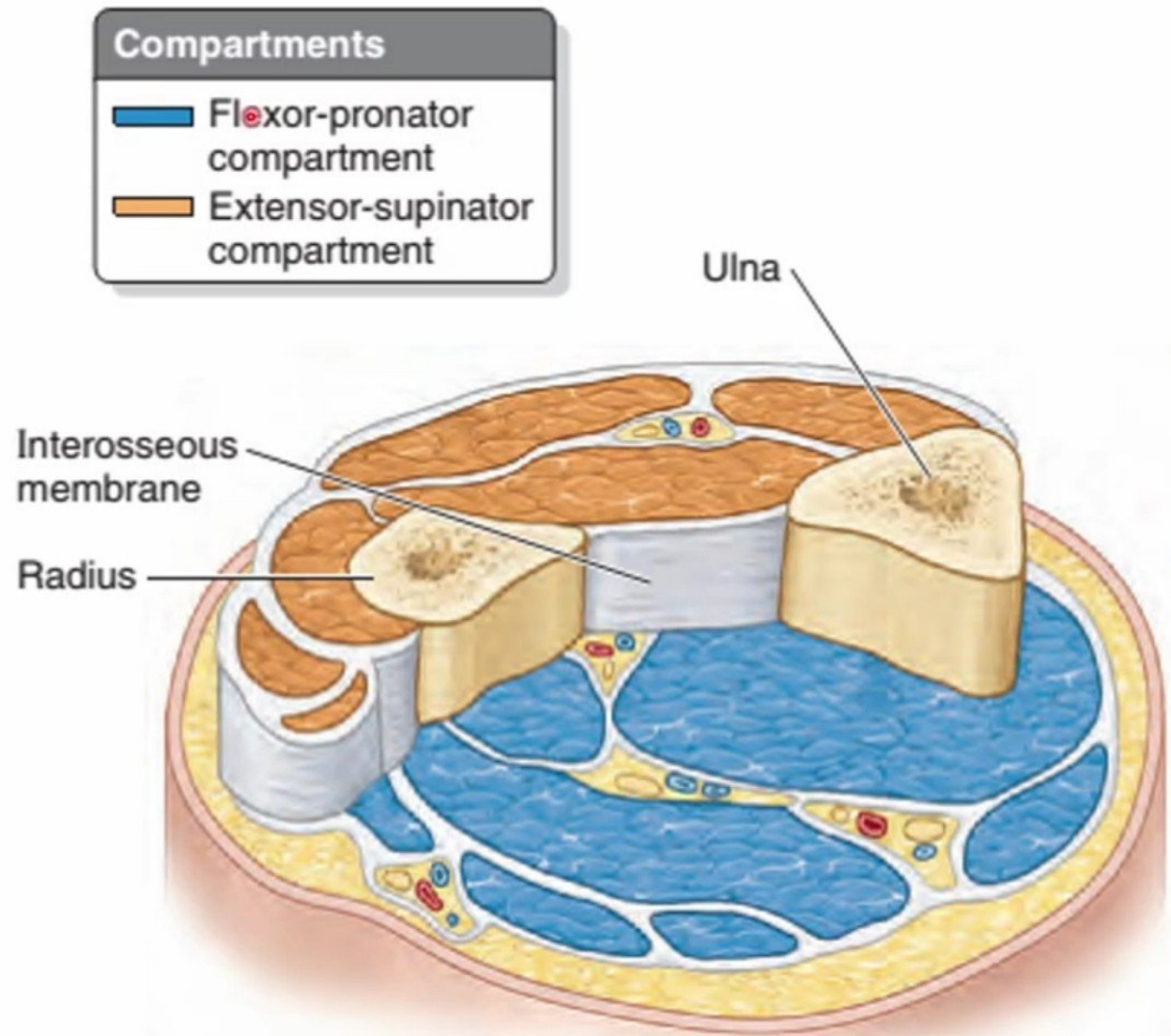
Introduction

- Forearm is the area between **the elbow** and **wrist joint**.
- Communicate **proximally** with **cubital fossa** & distally with the hand through **carpal tunnel**.



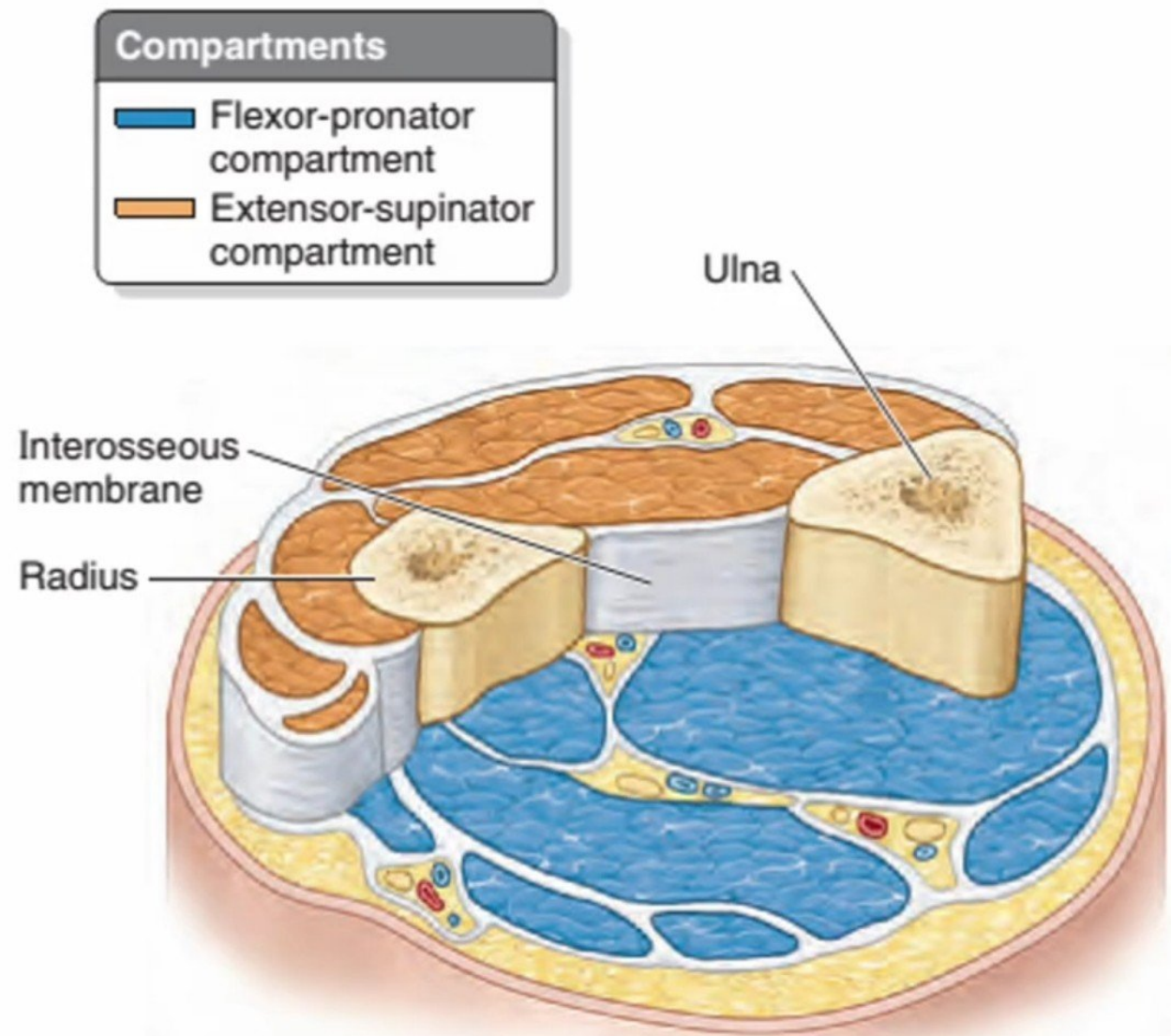
Compartments

- **Anterior:** contain muscles which move the wrist joint, flex the fingers & pronation.
- **Posterior:** contain muscles which move the wrist joint, extend the fingers & supination.
- *Each one has it own nerve, artery & deep veins.*



Compartments

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Muscles of the Forearm

Anterior compartments:

- Superficial
- Intermediate
- Deep

Posterior compartments:

- Superficial.
- Deep.

Contents of the Anterior Fascial Compartment of the Forearm

Muscles:

Superficial group consisting of:

- The pronator teres.
- The flexor carpi radialis.
- The palmaris longus.
- The flexor carpi ulnaris.

Intermediate group consisting of:

- The flexor digitorum superficialis.

Deep group consisting of :

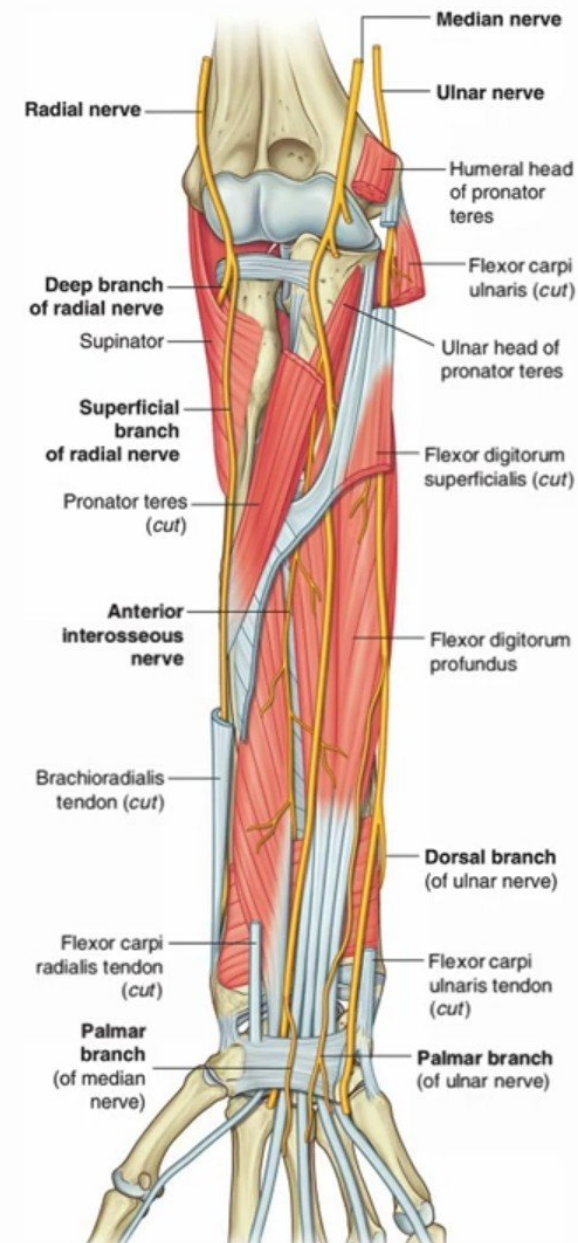
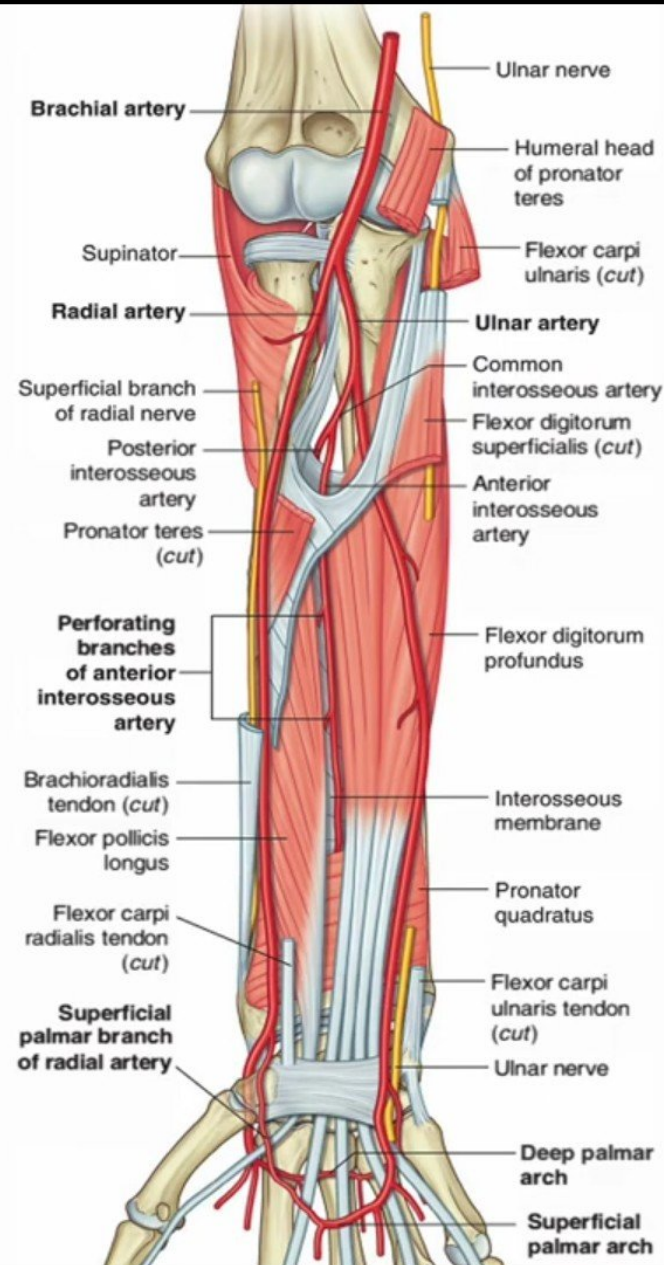
- The flexor pollicis longus.
- The flexor digitorum profundus.
- The pronator quadratus.

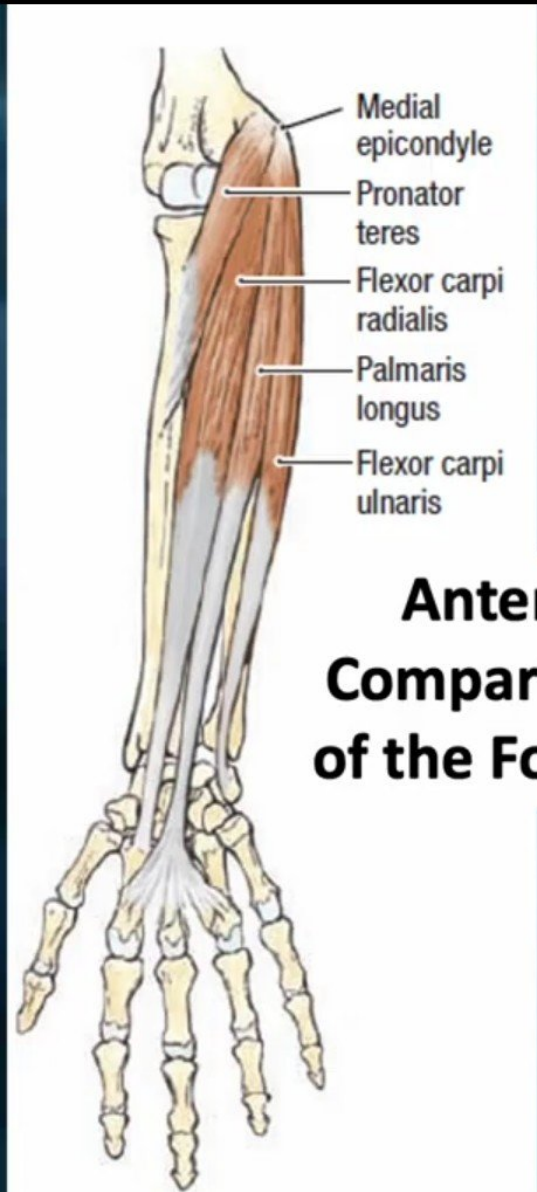
Blood supply to the muscles:

Ulnar and radial arteries.

Nerve supply to the muscles:

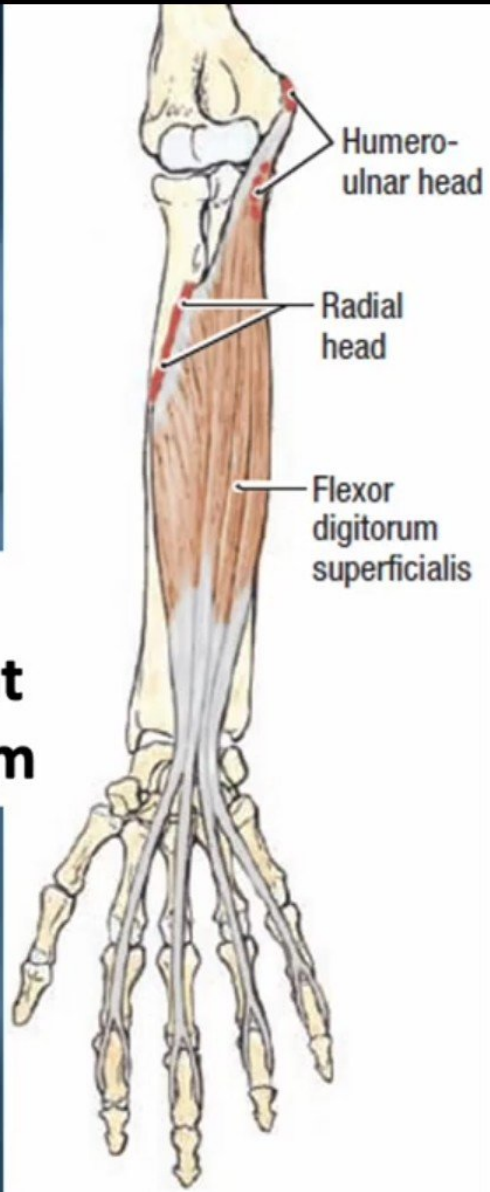
All the muscles are supplied by **the median nerve** and its branches, **except the flexor carpi ulnaris and the medial part of the flexor digitorum profundus**, which are supplied by the **ulnar nerve**.



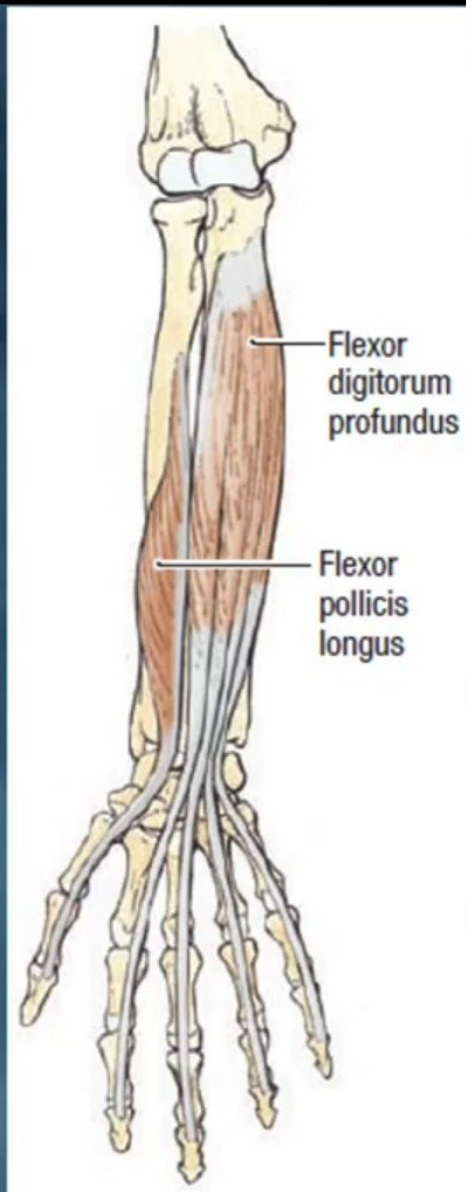


Anterior Compartment of the Forearm

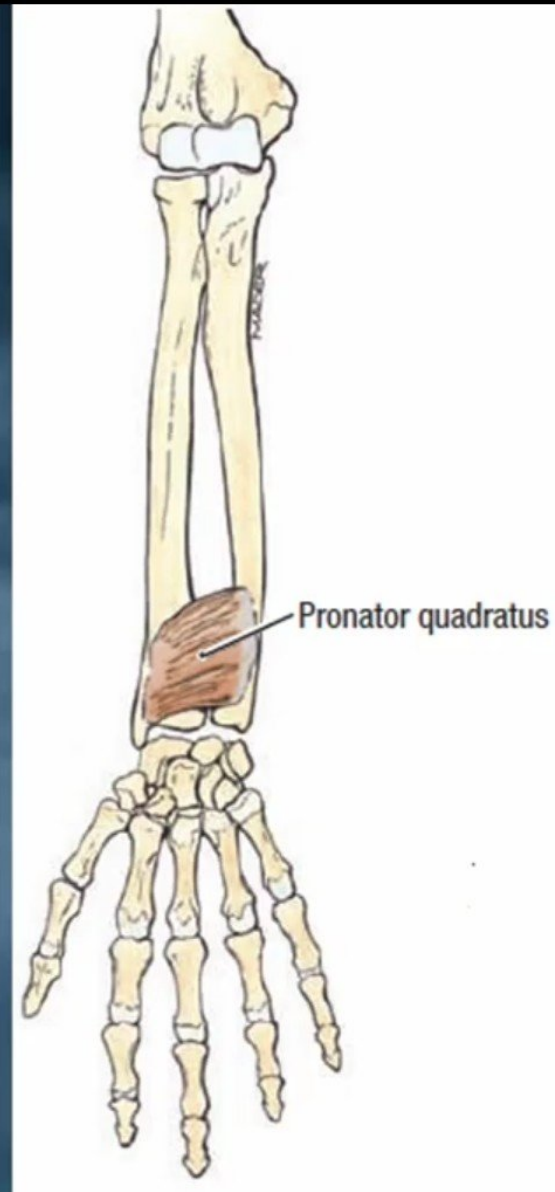
1st layer



2nd layer



3rd layer



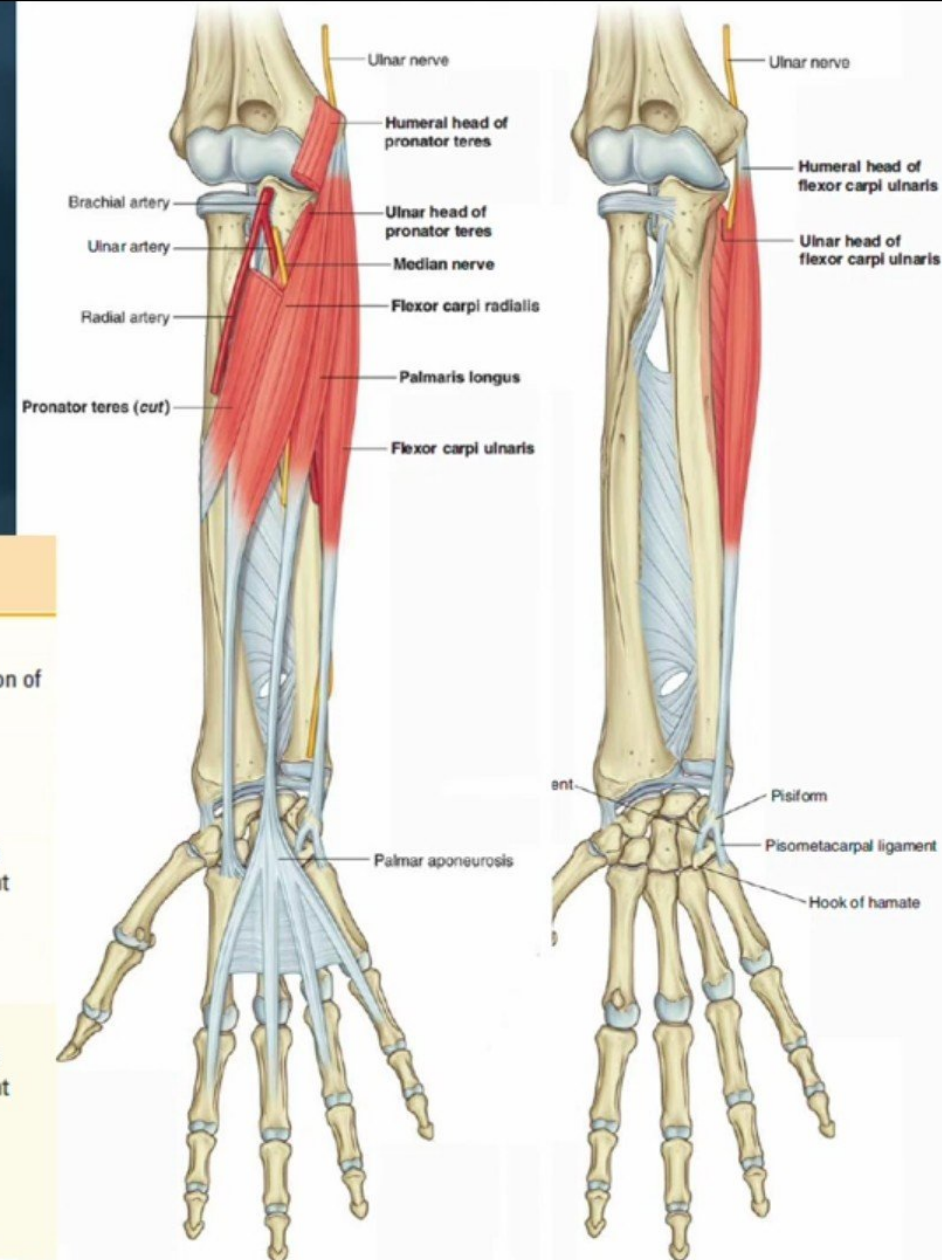
4th layer

Superficial layer

Muscles:

Superficial group consisting of:

- The pronator teres.
- The flexor carpi radialis.
- The palmaris longus.
- The flexor carpi ulnaris.



Muscle	Origin	Insertion	Nerve Supply	Nerve Roots ^a	Action
Pronator Teres					
Humeral head	Medial epicondyle of humerus	Lateral aspect of shaft of radius	Median nerve	C6, 7	Pronation and flexion of forearm
Ulnar head	Medial border of coronoid process of ulna				
Flexor carpi radialis	Medial epicondyle of humerus	Bases of second and third metacarpal bones	Median nerve	C6, 7	Flexes and abducts hand at wrist joint
Palmaris longus	Medial epicondyle of humerus	Flexor retinaculum and palmar aponeurosis	Median nerve	C7, 8	Flexes hand
Flexor Carpi Ulnaris					
Humeral head	Medial epicondyle of humerus	Pisiform bone, hook of the hamate, base at fifth metacarpal bone	Ulnar nerve	C8; T1	Flexes and adducts hand at wrist joint
Ulnar head	Medial aspect of olecranon process and posterior border of ulna				

Intermediate and Deep layers

Intermediate group consisting of:

➤ The flexor digitorum superficialis.

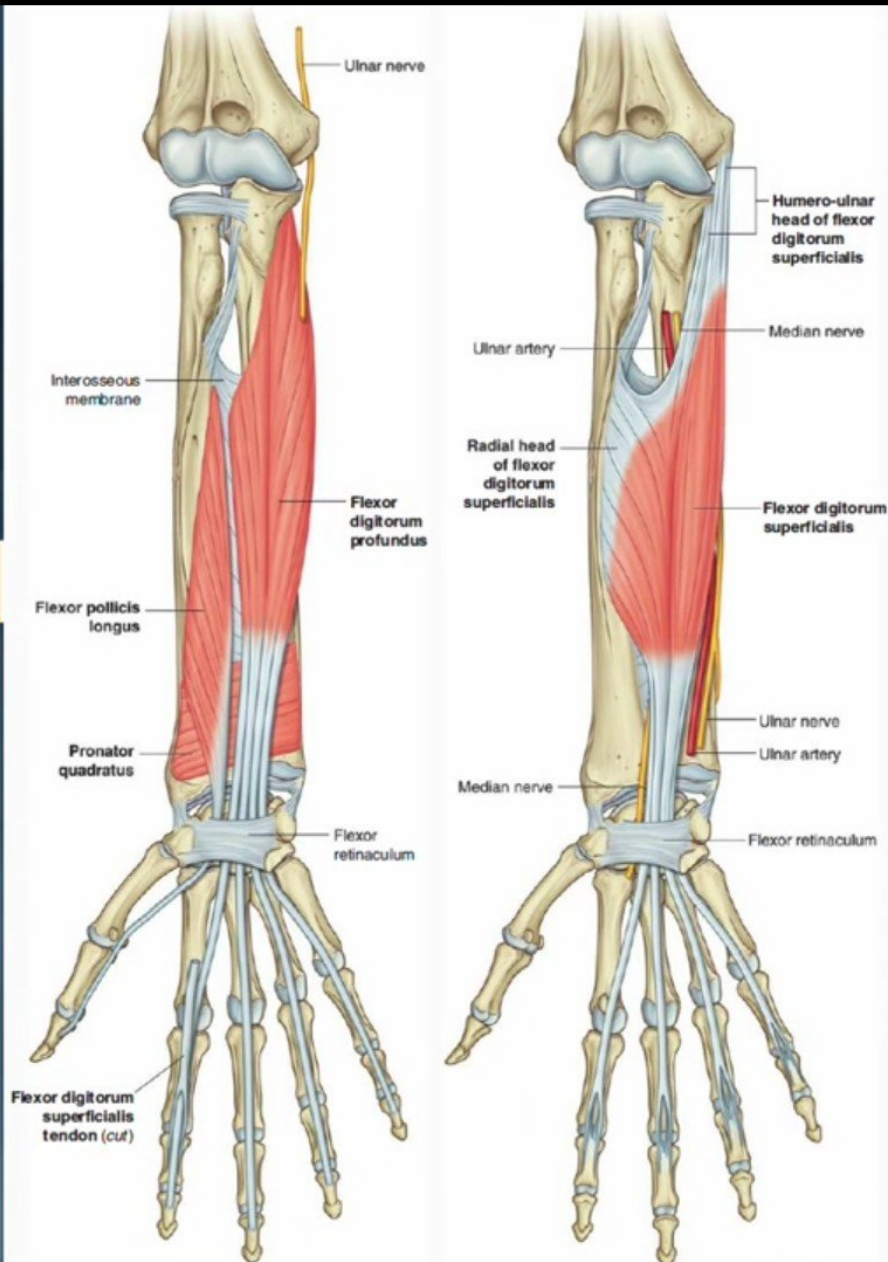
Deep group consisting of :

➤ The flexor pollicis longus.

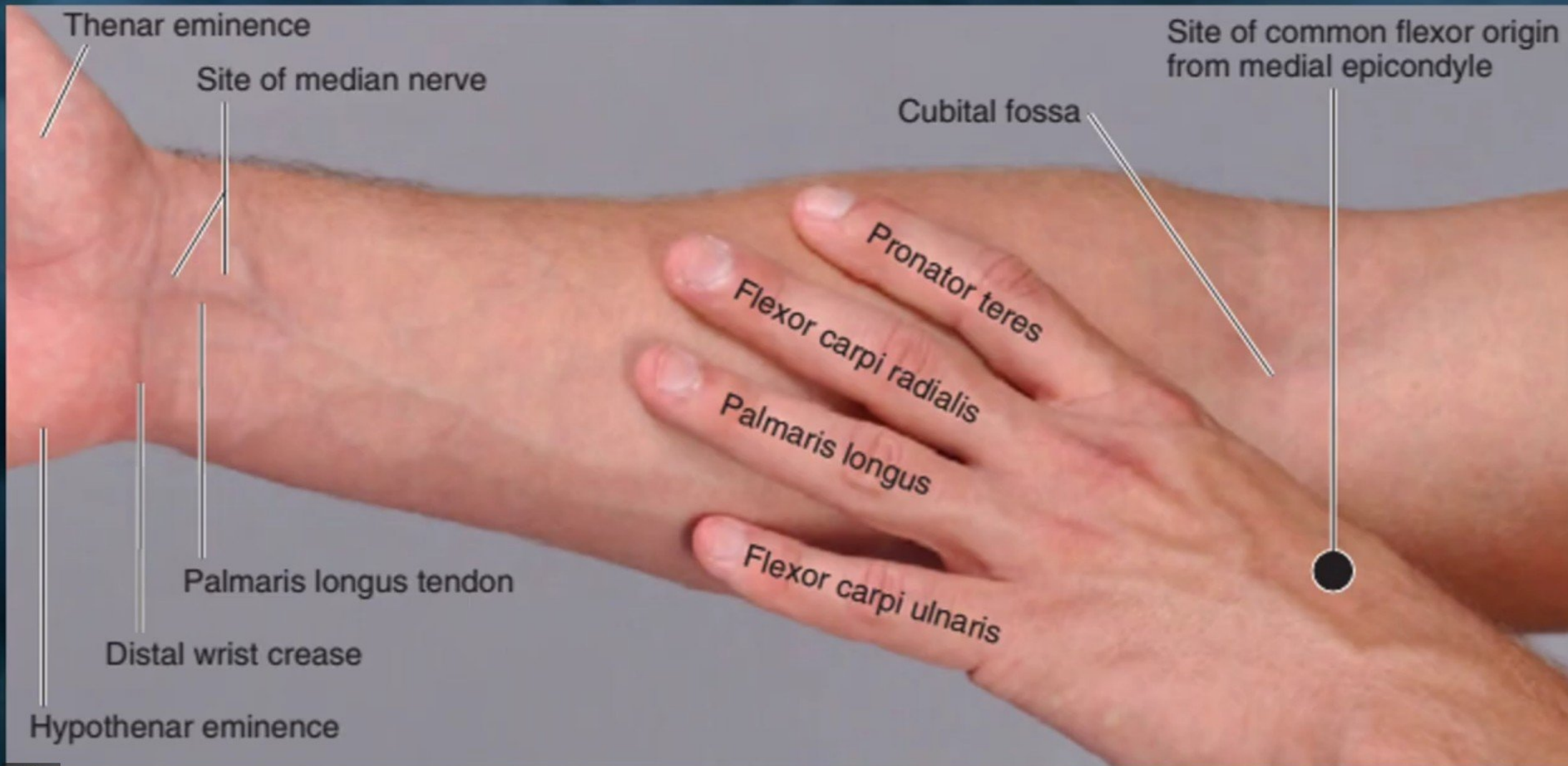
➤ The flexor digitorum profundus.

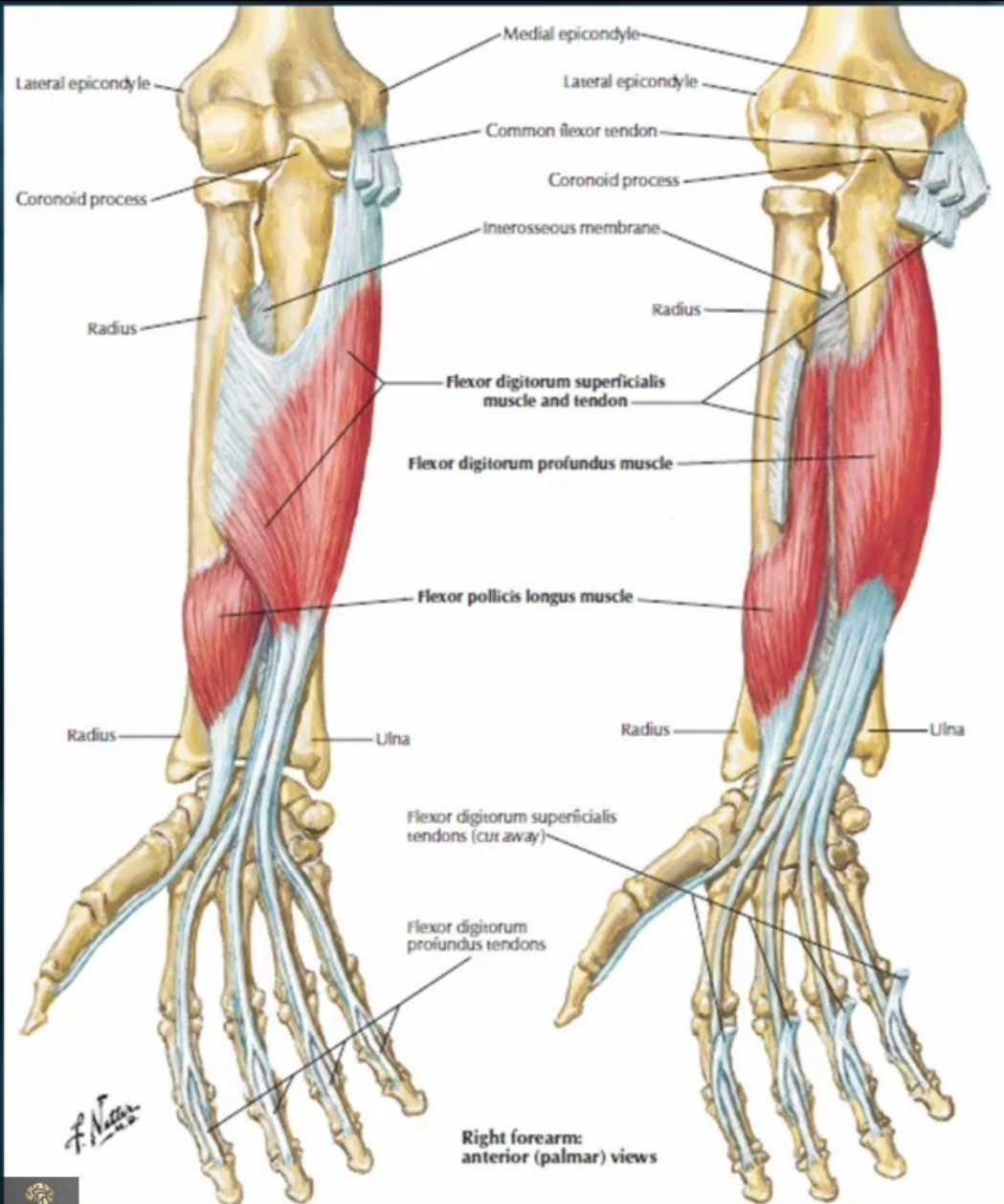
➤ The pronator quadratus.

Muscle	Origin	Insertion	Nerve Supply	Nerve Roots ^a	Action
Flexor Digitorum Superficialis					
Humero-ulnar head	Medial epicondyle of humerus; medial border of coronoid process of ulna	Middle phalanx of medial four fingers	Median nerve	C7, 8; T1	Flexes middle phalanx of fingers and assists in flexing proximal phalanx and hand
Radial head	Oblique line on anterior surface of shaft of radius				
Flexor pollicis longus	Anterior surface of shaft of radius	Distal phalanx of thumb	Anterior interosseous branch of median nerve	C8; T1	Flexes distal phalanx of thumb
Flexor digitorum profundus	Anteromedial surface of shaft of ulna	Distal phalanges of medial four fingers	Ulnar (medial half) and median (lateral half) nerves	C8; T1	Flexes distal phalanx of fingers; then assists in flexion of middle and proximal phalanges and wrist
Pronator quadratus	Anterior surface of shaft of ulna	Anterior surface of shaft of radius	Anterior interosseous branch of median nerve	C8; T1	Pronates forearm

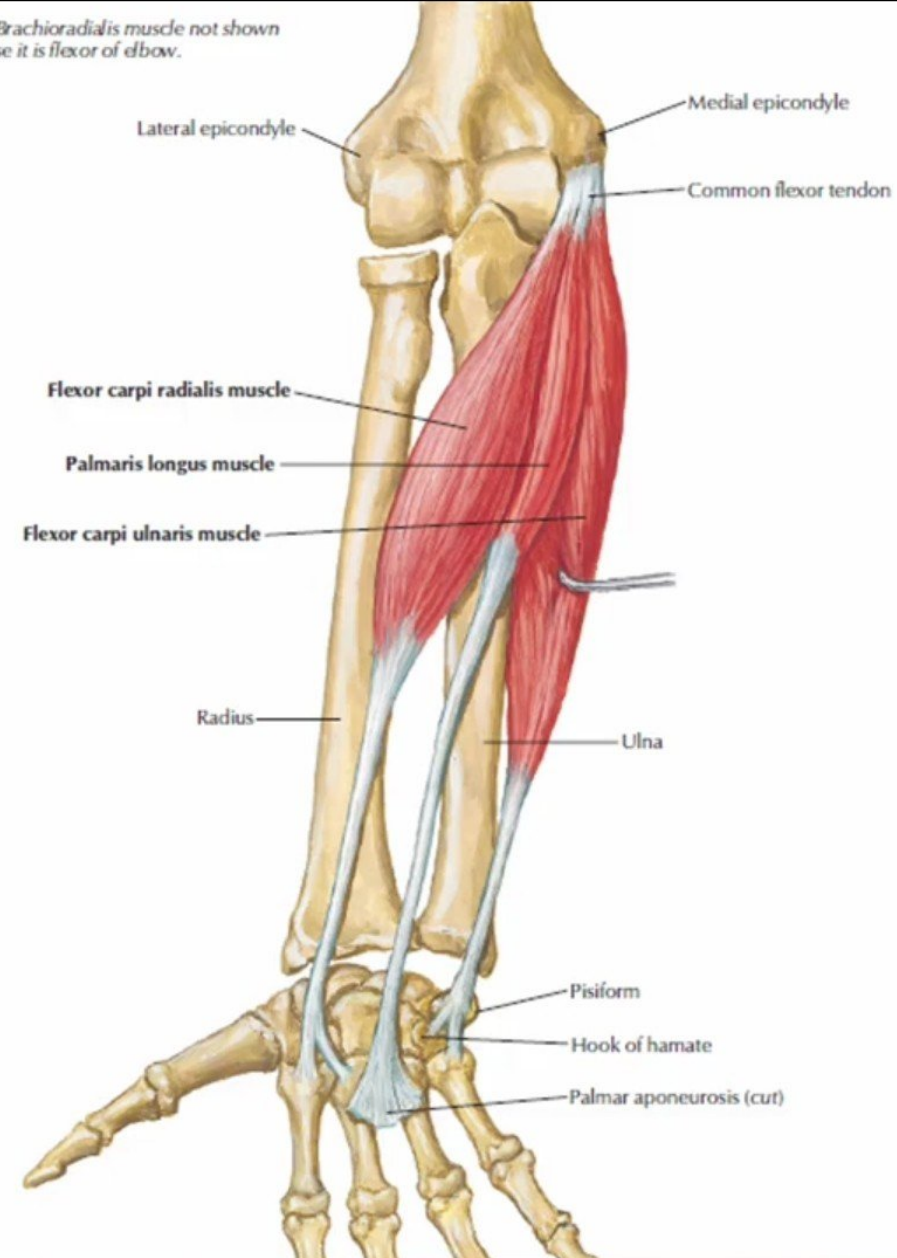


Fore remember ...





Note: Brachioradialis muscle not shown because it is flexor of elbow.



Contents of the Posterior Fascial Compartment of the Forearm

Muscles:

The superficial group includes:

- Brachioradialis.
- Extensor carpi radialis longus.
- Extensor carpi radialis brevis.
- Extensor digitorum.
- Extensor digiti minimi.
- Extensor carpi ulnaris.
- Anconeus.

These muscles possess a common tendon of origin (the lateral epicondyle of the humerus).

The deep group includes:

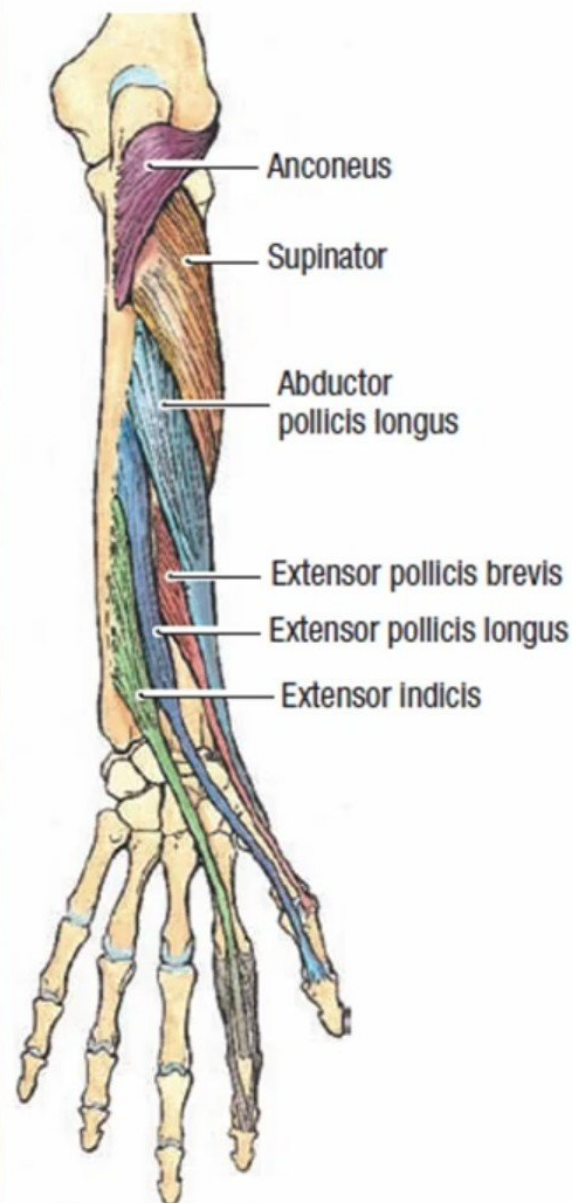
- Supinator
- Abductor pollicis longus,
- Extensor pollicis brevis,
- Extensor pollicis longus,
- Extensor indicis.

Blood supply:

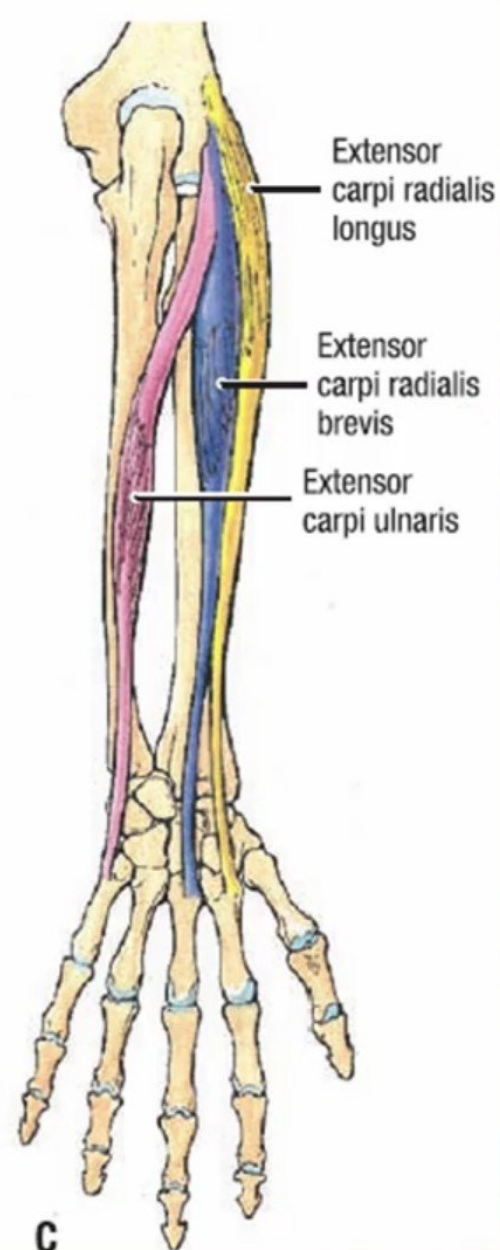
- Posterior and anterior interosseous arteries

Nerve supply:

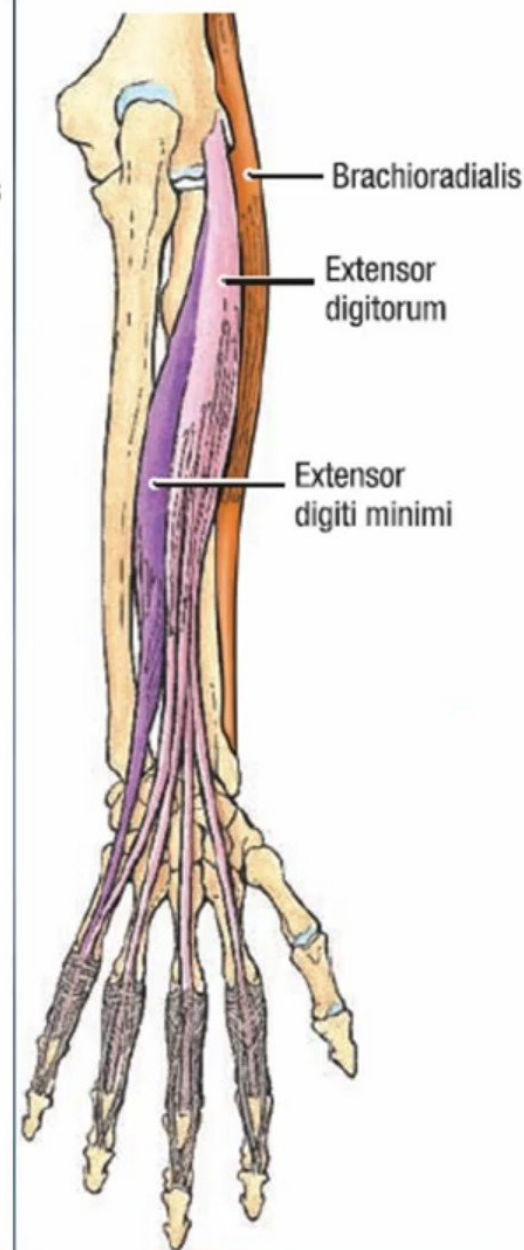
Deep branch of the radial nerve.



B. Posterior View

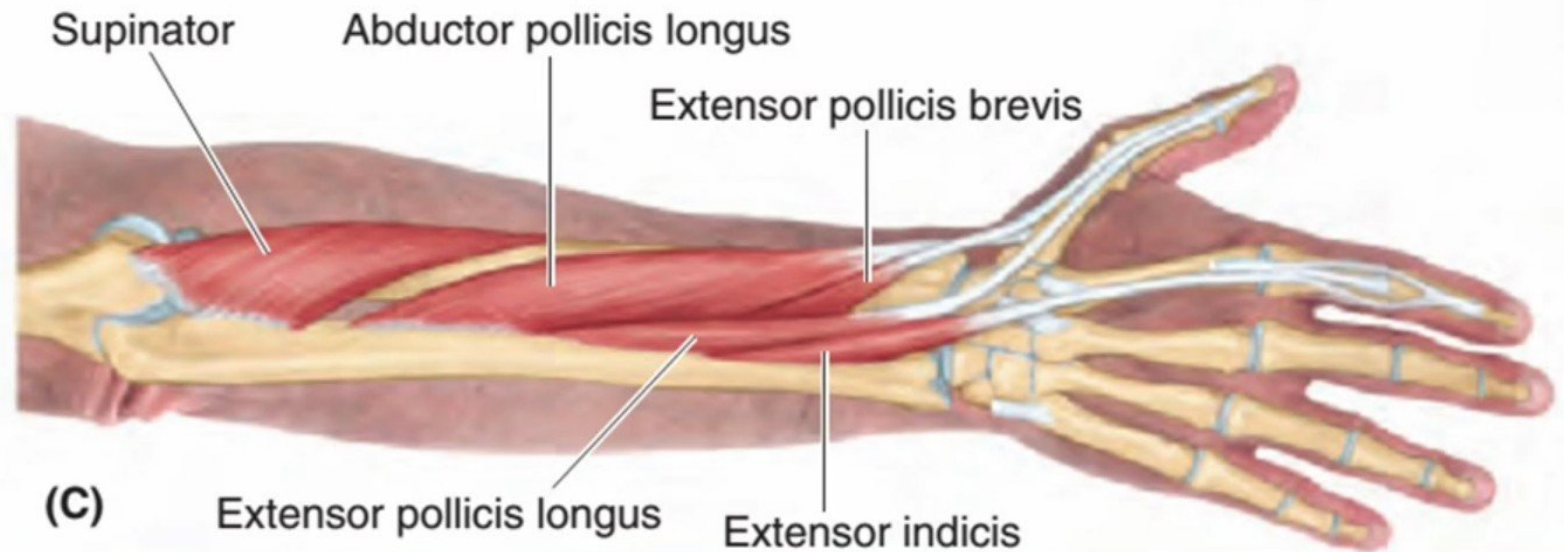
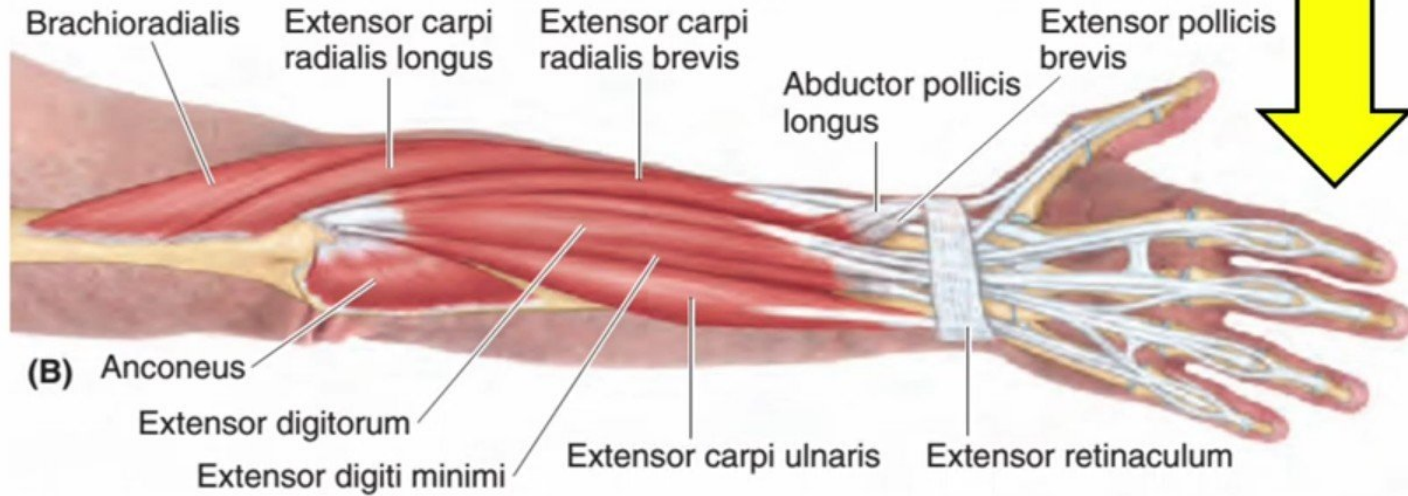
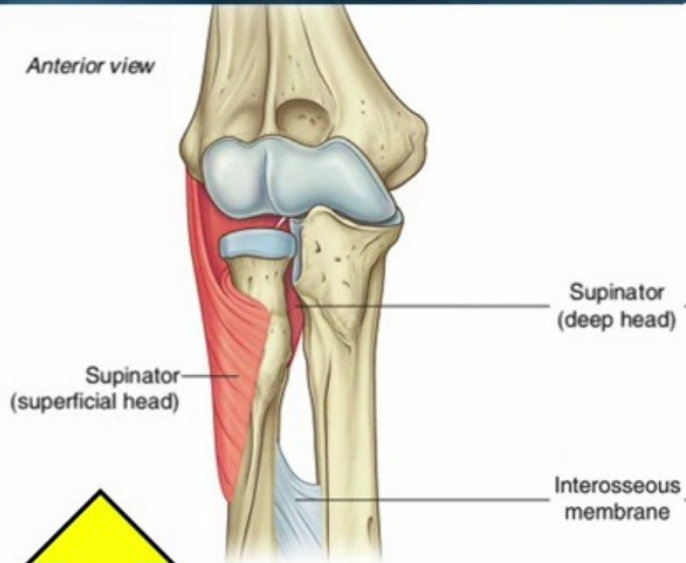


C



Posterior compartment

Superficial group

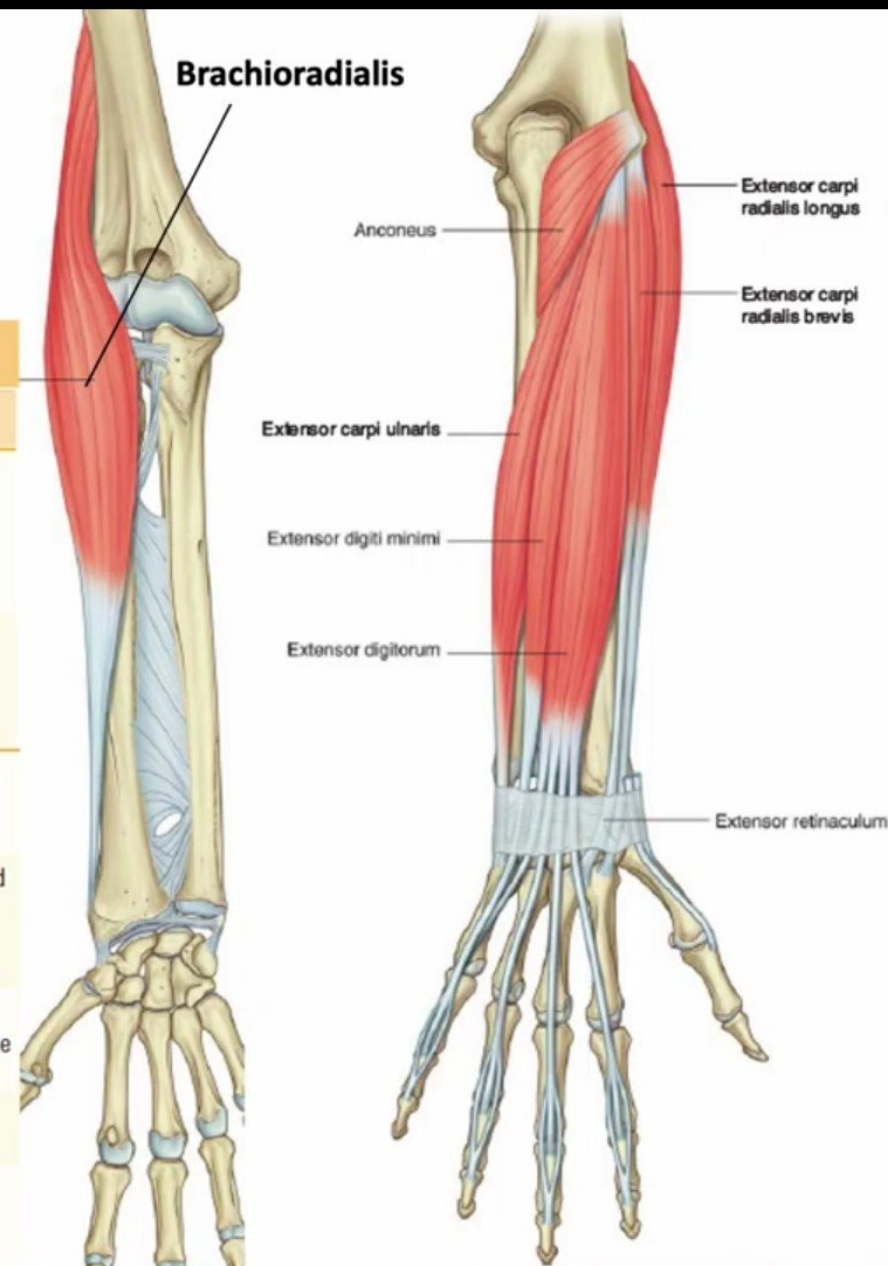


Deep
Group

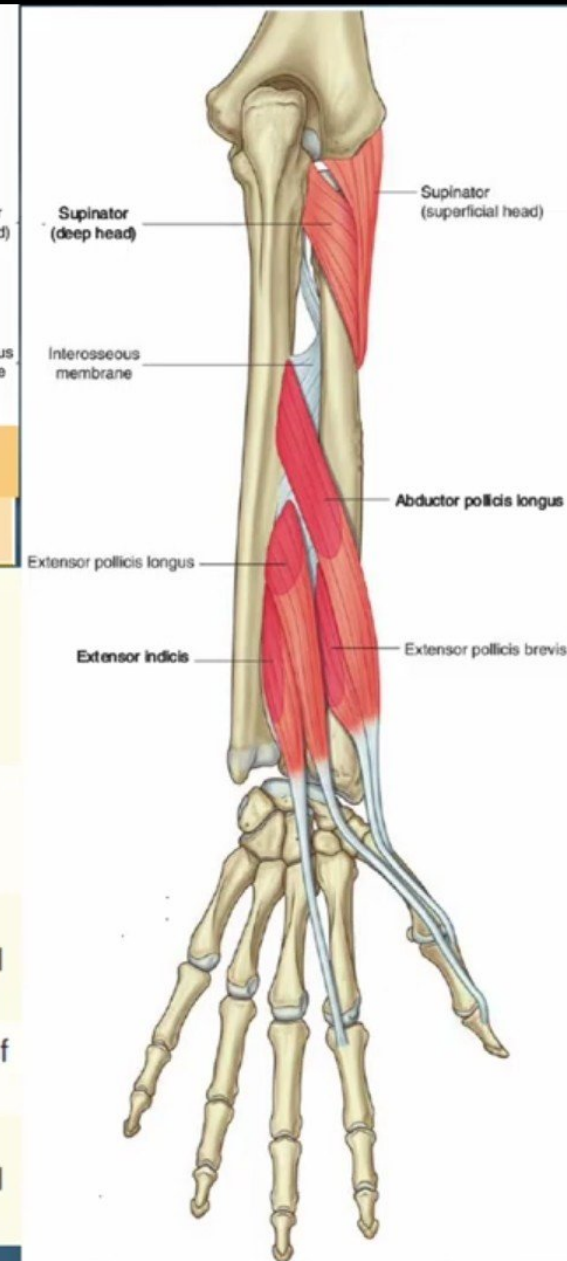
Superficial group

Muscles of the Posterior Fascial Compartment of the Forearm

Muscle	Origin	Insertion	Nerve Supply	Nerve Roots ^a	Action
Brachioradialis	Lateral supracondylar ridge of humerus	Base of styloid process of radius	Radial nerve	C5, 6, 7	Flexes forearm at elbow joint; rotates forearm to the midprone position
Extensor carpi radialis longus	Lateral supracondylar ridge of humerus	Posterior surface of base of second metacarpal bone	Radial nerve	C6, 7	Extends and abducts hand at wrist joint
Extensor carpi radialis brevis	Lateral epicondyle of humerus	Posterior surface of base of third metacarpal bone	Deep branch of radial nerve	C7, 8	Extends and abducts hand at wrist joint
Extensor digitorum	Lateral epicondyle of humerus	Middle and distal phalanges of medial four fingers	Deep branch of radial nerve	C7, 8	Extends fingers and hand (see text for details)
Extensor digiti minimi	Lateral epicondyle of humerus	Extensor expansion of little finger	Deep branch of radial nerve	C7, 8	Extends metacarpal phalangeal joint of little finger
Extensor carpi ulnaris	Lateral epicondyle of humerus	Base of 5th metacarpal bone	Deep branch of radial nerve	C7, 8	Extends and adducts hand at wrist joint
Anconeus	Lateral epicondyle of humerus	Lateral surface of olecranon process of ulna	Radial nerve	C7, 8; T1	Extends elbow joint



Deep Group



Muscles of the Posterior Fascial Compartment of the Forearm

Muscle	Origin	Insertion	Nerve Supply	Nerve Roots ^a	Action
Supinator	Lateral epicondyle of humerus, anular ligament of proximal radioulnar joint, and ulna	Neck and shaft of radius	Deep branch of radial nerve	C5, 6	Supination of forearm
Abductor pollicis longus	Posterior surface of shafts of radius and ulna	Base of first metacarpal bone	Deep branch of radial nerve	C7, 8	Abducts and extends thumb
Extensor pollicis brevis	Posterior surface of shaft of radius	Base of proximal phalanx of thumb	Deep branch of radial nerve	C7, 8	Extends metacarpophalangeal joints of thumb
Extensor pollicis longus	Posterior surface of shaft of ulna	Base of distal phalanx of thumb	Deep branch of radial nerve	C7, 8	Extends distal phalanx of thumb
Extensor indicis	Posterior surface of shaft of ulna	Extensor expansion of index finger	Deep branch of radial nerve	C7, 8	Extends metacarpophalangeal joint of index finger

*Will be continue.. &
good luck.. 😊*